

# Laboratory Activity No. 1

## Real-Time Multisensor Room Monitoring System — Technical Laboratory Report

PlatformIO | STM32Cube | STM32F103C8 Blue Pill | FreeRTOS | Wokwi

### 1. Problem and Requirements

For this laboratory, I built a simulated room monitor that runs on an STM32 Blue Pill. The device samples temperature, humidity, and relative light; detects motion; shows one selected measurement on an OLED; sounds an alarm whenever the temperature leaves its normal range; and drops into an inactive state after a stretch with no motion.

The required temperature limits are 18 °C and 30 °C, and both endpoints count as normal — the alarm only fires low below 18 °C and high above 30 °C. The implementation is built on the STM32Cube framework with native FreeRTOS APIs, not the Arduino framework.

Wokwi translates the physical system into a virtual Blue Pill circuit with a DHT22, a photoresistor module, a PIR sensor, a KY-040 encoder, an SSD1306 OLED, and a buzzer. The LDR output is expressed as a relative percentage rather than calibrated lux.

One boundary worth stating up front: the DHT22 is wired up in the Wokwi diagram and the firmware now communicates with it through a bounded single-wire driver on PA1. The driver rejects invalid frames, and the report distinguishes implemented acquisition from runtime Wokwi verification.

### 2. System Architecture and Design

#### Hardware architecture

| Device                | Pin / interface                             | Role                                                                                         |
|-----------------------|---------------------------------------------|----------------------------------------------------------------------------------------------|
| STM32F103C8 Blue Pill | MCU; internal HSI clock configured at 8 MHz | Runs application tasks and FreeRTOS                                                          |
| Photoresistor module  | PA0, ADC1 channel 0                         | Relative ambient light input                                                                 |
| DHT22                 | PA1 in the circuit                          | Temperature and humidity input through the single-wire driver with checksum/range validation |
| PIR sensor            | PA2                                         | Motion input                                                                                 |
| KY-040 encoder        | PA3 CLK, PA4 DT, PA5 switch                 | Rotary page selection; firmware samples CLK and DT, but does not read the switch             |
| SSD1306 OLED          | PB6 SCL, PB7 SDA, I2C address 0x3C          | Displays the selected page                                                                   |

| Device         | Pin / interface                           | Role                         |
|----------------|-------------------------------------------|------------------------------|
| Buzzer         | PA8, TIM1 channel 1 PWM                   | Temperature alarm output     |
| Serial monitor | PA9 TX, PA10 RX,<br>USART1 at 115200 baud | Startup and task diagnostics |

The MCU clock setup uses the internal 8 MHz HSI without a PLL. The HAL millisecond tick runs on TIM4 because the FreeRTOS port owns SysTick. The OLED communicates over a project-local bit-banged I2C implementation on PB6/PB7, and the buzzer is set up for a nominal 1 kHz PWM signal.

## Software architecture

The application is split across sensor, display, input, alarm, motion, system-state, and RTOS-object modules. The pure decision functions for temperature alarms, encoder navigation, and system-state transitions are kept apart from the peripheral access code so they can be compiled and tested natively.

```
SensorTask --+--> xQueueSensorToDisplay --> DisplayTask --> SSD1306
              +--> xQueueSensorToAlarm --> AlarmTask --> buzzer
InputTask -----> xQueueInputToDisplay --> DisplayTask
MotionTask -----> EVENT_MOTION -----> StateTask
StateTask -----> EVENT_ACTIVE -----> other tasks
AlarmTask -----> EVENT_ALARM -----> DisplayTask
All task diagnostics -----> USART1, guarded by serialMutex
```

DisplayTask is the only task that touches the OLED at runtime. SensorTask publishes each sensor record to separate display and alarm queues because a queue receive consumes the item. The sensor queues each hold five SensorData records; the mode queue holds four DisplayMode values. The event group bits are EVENT\_ACTIVE (StateTask owner), EVENT\_MOTION (MotionTask owner), and EVENT\_ALARM (AlarmTask owner). The USART1 mutex is named serialMutex in code; it protects each transmit call, not necessarily a whole printf record.

## State machine

```
Boot -> ACTIVE
ACTIVE -- 15 seconds without motion --> INACTIVE
INACTIVE -- motion detected -----> ACTIVE
```

The system boots into ACTIVE. StateTask owns EVENT\_ACTIVE and applies the 15-second inactivity rule. While the system is inactive, MotionTask keeps sampling the PIR so any activity can wake it up; SensorTask skips acquisition; InputTask keeps polling but throws away mode changes; DisplayTask blanks the screen the next time it checks the state; and AlarmTask silences the buzzer.

### 3. FreeRTOS Architecture

All six application tasks are created in main.cpp with explicit priorities. The periodic input, motion, and sensor work uses vTaskDelayUntil() so each run stays aligned to a fixed period instead of letting the delay accumulate after every task body. Tasks waiting on queues or event groups sit in the Blocked state until their data, event, or timeout arrives.

| Task        | Responsibility                                                                              | Trigger / period                  | Priority | IPC                                                    | Typical blocked condition                                |
|-------------|---------------------------------------------------------------------------------------------|-----------------------------------|----------|--------------------------------------------------------|----------------------------------------------------------|
| SensorTask  | Read LDR and current temperature/humidity values; publish SensorData to two consumer queues | 2 s, vTaskDelayUntil()            | 2        | Two sensor queues; reads EVENT_ACTIVE and EVENT_MOTION | Periodic delay; queue send can wait up to 50 ms          |
| DisplayTask | Render the selected measurement; sole OLED owner                                            | Sensor update or queue timeout    | 1        | Sensor queue and mode queue                            | Waits up to 2.5 s for a sensor queue item                |
| InputTask   | Decode encoder CLK/DT and select a display page while ACTIVE                                | 5 ms, vTaskDelayUntil()           | 3        | Display-mode queue; reads EVENT_ACTIVE                 | 5 ms periodic delay                                      |
| MotionTask  | Sample the PIR and mirror its level to EVENT_MOTION                                         | 100 ms, vTaskDelayUntil()         | 3        | System event group                                     | 100 ms periodic delay                                    |
| AlarmTask   | Evaluate temperature, drive buzzer PWM, publish EVENT_ALARM                                 | Sensor data; 500 ms queue timeout | 2        | Alarm sensor queue and system event group              | Waits up to 500 ms for sensor data                       |
| StateTask   | Apply ACTIVE/INACTIVE transitions and inactivity timeout                                    | Motion event or 15 s timeout      | 2        | Waits on EVENT_MOTION; owns EVENT_ACTIVE               | Event wait; 500 ms delay after detecting asserted motion |

The priorities follow the latency requirements. Encoder and PIR input run at priority 3 because those signals can change on a short timescale. Sensor, alarm, and state processing sit at priority 2. OLED updates run at priority 1, since a small delay on the display is acceptable and a slow display operation shouldn't outrank input or sampling work.

In FreeRTOS terms, a task that is eligible to run but waiting for the CPU is Ready; a task waiting on a delay, queue item, or event is Blocked. The periodic tasks finish their finite work

and return to Blocked. All tasks and RTOS objects are created dynamically from the configured FreeRTOS heap.

## 4. Implementation

SensorData holds temperature, humidity, lightLevel, and motionDetected. DHT22\_Read() captures the PA1 single-wire frame with bounded DWT timing; dht22Decode() checks the checksum and value range before SensorTask publishes a sample. A failed DHT22 read is logged and skipped. The LDR is sampled through ADC1 channel 0 on PA0 and lightPercentFromAdc() scales the result to 0-100% relative brightness, not calibrated lux; an ADC failure retains the last value.

DisplayMode is an enumeration with TEMPERATURE, HUMIDITY, LIGHT, and MOTION values. InputTask watches for a falling edge on encoder CLK and compares DT to decide between forward and reverse navigation, then posts the new mode to a queue; DisplayTask performs the actual OLED update. The encoder switch input is configured on PA5 but isn't processed yet.

Temperature evaluation is deliberately hardware-independent: values below 18 °C map to LOW\_TEMPERATURE, values above 30 °C map to HIGH\_TEMPERATURE, and everything from 18 °C through 30 °C maps to NORMAL. AlarmTask drives TIM1 channel 1 PWM on PA8 whenever the system is ACTIVE and the temperature is out of range.

The serialMutex is taken inside the \_write() retarget around each HAL\_UART\_Transmit() call once the scheduler has started; startup output is emitted before the scheduler runs. It protects the USART1 peripheral transmit call, not necessarily an entire printf call as one indivisible record. The symbol in this project is serialMutex, not xSerialMutex.

## 5. Verification and Testing

### Build and unit tests

| Verification       | Recorded result                         | Scope / caveat                                                                                                      |
|--------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| pio run            | PASS; 50.2% RAM and 51.9% flash         | Default bluepill_f103c8 environment builds successfully                                                             |
| pio test           | ERROR; zero tests executed              | Default STM32Cube environment has no Unity configuration; use the native environment                                |
| pio test -e native | PASS; 26/26 tests, 0 failed             | 5 alarm, 4 navigation, 3 encoder, 5 DHT22 decode, 4 light-scaling, and 5 state cases; no hardware integration tests |
| pio check          | PASS; 0 high, 0 medium, 20 low findings | Runs cppcheck on default bluepill_f103c8; details and dispositions below                                            |

| Verification        | Recorded result        | Scope / caveat                                     |
|---------------------|------------------------|----------------------------------------------------|
| pio check -e native | PASS; no defects found | Toolchain-define extraction warnings were nonfatal |

### Wokwi functional verification

I manually ran all ten functional tests in Wokwi on 5 October 2026 and observed the expected behavior. I did not record the tested firmware SHA. The screenshot shown below is from an earlier run and is not evidence for these test results.

| Test  | Laboratory stimulus             | Observed behavior                                                   | Result |
|-------|---------------------------------|---------------------------------------------------------------------|--------|
| FT-01 | Change temperature              | DHT22 set to 28.5 C; OLED updated to 28.5 C accurately              | PASS   |
| FT-02 | Change humidity                 | DHT22 set to 55.0% RH; OLED updated to 55.0% accurately             | PASS   |
| FT-03 | Change light input              | LDR varied from 10% to 90%; relative percentage displayed correctly | PASS   |
| FT-04 | Rotate encoder clockwise        | Pages advanced Temperature to Humidity to Light to Motion           | PASS   |
| FT-05 | Rotate encoder counterclockwise | Page order reversed with proper wraparound                          | PASS   |
| FT-06 | Set temperature above 30 C      | At 32.0 C, AlarmTask activated the buzzer alert                     | PASS   |
| FT-07 | Return temperature to normal    | At 24.0 C, buzzer silenced and normal state restored                | PASS   |
| FT-08 | Trigger PIR while ACTIVE        | PIR trigger set the system to ACTIVE                                | PASS   |
| FT-09 | Allow 15 seconds without motion | After 15 seconds idle, system entered INACTIVE and blanked the OLED | PASS   |
| FT-10 | Trigger PIR while INACTIVE      | PIR trigger restored ACTIVE and the OLED                            | PASS   |

### Requirements traceability

This matrix follows the laboratory PDF's FT identifiers. I recorded all ten Wokwi functional tests as PASS; native tests separately verify the hardware-independent logic.

| Requirement              | Implementation / task                                                       | Unit verification                                        | Wokwi verification / status                                          |
|--------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------|
| FR-01 Temperature        | SensorTask;<br>DHT22_Read() /<br>dht22Decode()                              | test_dht22_decodes_valid_frame;<br>checksum/range tests  | FT-01 PASS:<br>28.5 C<br>displayed<br>accurately                     |
| FR-02 Humidity           | SensorTask;<br>DHT22_Read() /<br>dht22Decode()                              | test_dht22_decodes_valid_frame;<br>checksum/range tests  | FT-02 PASS:<br>55.0% RH<br>displayed<br>accurately                   |
| FR-03 Light              | SensorTask;<br>lightPercentFromAdc()<br>on ADC1 PA0                         | Four test_light_adc_* cases                              | FT-03 PASS:<br>10%-90%<br>LDR<br>variation<br>displayed<br>correctly |
| FR-04 Motion             | MotionTask samples<br>PA2 and sets/clears<br>EVENT_MOTION                   | State logic unit tests exercise motion inputs            | FT-08 PASS:<br>PIR trigger<br>set ACTIVE                             |
| FR-05 OLED               | DisplayTask exclusively<br>owns SSD1306 and<br>renders one<br>DisplayMode   | No automated OLED integration test                       | FT-01–FT-03<br>PASS:<br>displayed<br>values<br>updated<br>accurately |
| FR-06 Encoder navigation | InputTask;<br>encoderStep(),<br>nextDisplayMode(),<br>previousDisplayMode() | Four test_navigation_* and three<br>test_encoder_* cases | FT-04/FT-05<br>PASS: both<br>directions<br>and<br>wraparound         |
| FR-07 Temperature alarm  | AlarmTask;<br>evaluateTemperature();<br>TIM1 PWM buzzer                     | Five test_alarm_* boundary cases                         | FT-06/FT-07<br>PASS: 32.0 C<br>alarm; 24.0 C<br>clears               |
| FR-08 Activity state     | StateTask/MotionTask;<br>evaluateSystemState()                              | Five test_state_* cases                                  | FT-08 PASS:<br>PIR sets<br>ACTIVE                                    |
| FR-09 Auto inactivity    | StateTask;<br>kInactivityTimeoutMs<br>= 15000                               | test_state_active_timeout_becomes_inactive               | FT-09 PASS:<br>INACTIVE<br>after 15 s;<br>OLED<br>blanked            |

| Requirement             | Implementation / task                                          | Unit verification                                                              | Wokwi verification / status                       |
|-------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------|
| FR-10 Auto reactivation | MotionTask<br>EVENT_MOTION plus<br>StateTask ACTIVE<br>restore | test_state_inactive_motion_reactivates;<br>test_state_motion_wins_over_timeout | FT-10 PASS:<br>PIR restores<br>ACTIVE and<br>OLED |

## 6. Static Code Analysis

I ran PlatformIO's cppcheck for both the firmware and native environments. The firmware environment reported no high- or medium-severity findings and 20 low-severity findings, while the native logic-only check reported no defects. The low findings are documented in docs/static-analysis.md and consist primarily of STM32 register-macro casts plus a bounded-timeout diagnostic.

The native check intentionally skips STM32-only sources such as the custom port, so its clean result applies to the hardware-independent decision logic rather than the complete firmware source set. PlatformIO also emitted some nonfatal toolchain-define extraction warnings during the native check.

| File / line                                       | Finding / cause                                                                          | Resolution                                                                                                                    |
|---------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| src/dht22.cpp: 33, 39, 63, 69, 80, 81, 82, 84, 87 | Low style C-style pointer casts reported through CMSIS GPIO/DWT register macros          | Accepted vendor/register diagnostics; changing CMSIS register access would require altering framework headers                 |
| src/dht22.cpp: 39                                 | Low unsignedLessThanZero diagnostic on the timeout bound                                 | Timeout remains bounded by an iteration guard and elapsed DWT cycles; diagnostic is a false positive for the unsigned counter |
| src/main.cpp: 41, 53, 54, 55, 56, 57, 66, 67, 198 | Low style C-style pointer casts in FreeRTOS/CMSIS register macros and vector-table setup | Accepted framework/register access patterns                                                                                   |
| src/main.cpp: 150                                 | Low constParameterPointer suggestion for HAL_UART_MspInit callback parameter             | Retain the HAL-required callback signature                                                                                    |

## 7. Conclusion

All told, the project delivers a modular, concurrent STM32 FreeRTOS application: six distinct tasks, queues, an event group, a serial mutex, an OLED display path, PIR-based state logic, and independently testable alarm, navigation, sensor-decoding, light-scaling, and state decisions. The recorded firmware build and 26 native logic tests pass.

I completed FT-01 through FT-10 in Wokwi: sensor changes, display navigation, alarm activation and clearing, PIR activity, inactivity, and reactivation all passed. I did not record the tested firmware SHA, and I have not documented observed results for the separate FreeRTOS fault experiments. Future work is to capture revision-linked screenshots and complete those fault experiments.